Vitaran Edutech

# HINTS & SOLUTIONS

# INTEGRATED PROGRAM

# Biology

Class – 9

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# 1. The Fundamental Unit of Life-Cell

## Solutions

### Worksheet – 1

1. **Correct Answer: B**
   **Micrographia** – Robert Hooke published this book in 1665, using the word "cell" to describe compartments in cork.
2. **Correct Answer: C**
   **Rudolf Virchow** – He added this specific tenet to the cell theory in 1855, emphasizing cellular continuity.
3. **Correct Answer: C**
   **Cell wall** – While both have membranes, the rigid outer cell wall is a defining feature of plant cells, fungi, and bacteria.
4. **Correct Answer: C**
   **Centrosome –** The text notes that centrosomes (containing centrioles) are present in animal cells but absent in plant cells.
5. **Correct Answer: B**

   **Cell** – A cell is the fundamental structural and functional unit capable of independent existence and life functions.
6. **Correct Answer: B**

   **Multicellular** – These organisms, like humans and plants, consist of millions of cells performing specialized functions.
7. **Correct Answer: A**

   **1665** – Robert Hooke first observed the "small rooms" in cork tissue and coined the term in this year.
8. **Correct Answer: A**

   **Plastids** – Plastids (like chloroplasts) are specialized organelles for food synthesis/storage found in plants, not animals.
9. **Correct Answer: D**

   **Virchow** – This Latin phrase translates to "all cells arise from cells," the principle established by Rudolf Virchow.
10. **Correct Answer: B**
    **Activity** – An organism's overall activity is determined by the specific way its individual cell's function.

### Worksheet – 2

1. **Correct Answer: B**

   Cytology is the branch of biology specifically dedicated to the study of cells in terms of structure, function, and chemistry.
2. **Correct Answer: C**

   **Different components like lipids and proteins** the model describes the membrane as a "mosaic" because it is a flexible layer made of various molecules, primarily phospholipids and embedded proteins.
3. **Correct Answer: B**

   Phosphoglycerides (a type of phospholipid) are the main structural components that form the lipid bilayer of the cell membrane.
4. **Correct Answer: C**

   Osmosis is the specific term for the diffusion of water through a semi-permeable membrane from a region of high-water concentration to low.
5. **Correct Answer: C**

   **Against concentration gradient** Active transport requires energy (ATP) because it pumps molecules "uphill" from an area of low concentration to an area of high concentration.
6. **Correct Answer: A**

   Endocytosis is a cellular process in which substances are brought into the cell by the infolding of the cell membrane to form a vesicle.

7. **Correct Answer: B**
Pinocytosis is a form of endocytosis where the cell takes in fluids and dissolved small molecules; the name literally translates to "cell drinking."
8. **Correct Answer: C**
**Remain unchanged** in an isotonic solution, the concentration of solutes is equal inside and outside the cell, resulting in no net movement of water.
9. **Correct Answer: A**
**Into** In a hypotonic solution, the surrounding fluid has a lower solute concentration than the cell, causing water to rush inside to balance it out.
10. **Correct Answer: B**
Phagocytosis is a type of endocytosis used by immune cells like neutrophils to "eat" and digest large particles like bacteria or dead cells.

## Worksheet 3

1. **Correct Answer: B**
**Most abundant natural biopolymer** Cellulose is the primary structural component of plant cell walls and is the most widely occurring organic compound on Earth.
2. **Correct Answer: B**
Plasmodesmata are microscopic channels that traverse the cell walls of plant cells, enabling transport and communication between them.
3. **Correct Answer: A**
**Adhesion** The middle lamella is a layer rich in pectin that acts as a glue to bind the cell walls of adjacent plant cells together.
4. **Correct Answer: A**
**Nucleus and cytoplasm** Protoplasm is the living content of a cell that is surrounded by the plasma membrane, traditionally composed of the cytoplasm and the nucleus.
5. **Correct Answer: C**
**Multinucleate** Cells that contain multiple nuclei within a single plasma membrane are termed multinucleate (common in muscle fibers or certain fungi).
6. **Correct Answer: A**
**Lipids** The Smooth Endoplasmic Reticulum (SER) is responsible for the synthesis of lipids, phospholipids, and steroid hormones.
7. **Correct Answer: B**
**ATP** Mitochondria perform cellular respiration to convert nutrients into adenosine triphosphate (ATP), the primary energy currency of the cell.
8. **Correct Answer: C**
**70S** Prokaryotic cells contain 70S ribosomes (made of 50S and 30S subunits), which are smaller than the 80S ribosomes found in eukaryotic cytoplasm.
9. **Correct Answer: D**
The trans face (or maturing face) is the exit side of the Golgi apparatus where modified proteins are packaged into vesicles and sent to their destinations.
10. **Correct Answer: D**
**Excretion of excess water** Contractile vacuoles act as an osmoregulatory organelle that collects and pumps out excess water to prevent the *Amoeba* from bursting.

## Worksheet 4

1. **Correct Answer: A**
**Cytosomes** Microbodies are a class of small, membrane-bound organelles (like peroxisomes) that are also referred to as cytosomes.
2. **Correct Answer: D**
**Pulling chromatids apart** Centrioles help organize the spindle fibers that attach to and separate chromosomes during the process of cell division.
3. **Correct Answer: B**
**1–10 µm** Prokaryotic cells are significantly smaller than eukaryotic cells, typically measuring between 1 and 10 micrometers in diameter.
4. **Correct Answer: C**
**Linear chromosomes** Unlike the circular DNA found in prokaryotes, eukaryotic DNA is organized into multiple linear strands associated with histone proteins.

5. **Correct Answer: C**
**Latin** The word "centrum" is a Latin term meaning "middle" or "center point."
6. **Correct Answer: A**
**Prokaryotic** Mycoplasma are the smallest known free-living organisms and are classified as prokaryotes because they lack a nucleus and membrane-bound organelles.
7. **Correct Answer: C**
**Break down food vacuoles** Lysosomes contain hydrolytic enzymes that fuse with food vacuoles or damaged organelles to digest their contents.
8. **Correct Answer: D**
**Binary fission** Prokaryotes reproduce asexually through binary fission, a process where the cell duplicates its genetic material and divides into two equal parts.
9. **Correct Answer: B**
**Double membrane** the nucleus is enclosed by the nuclear envelope, which consists of two lipid bilayer membranes (inner and outer).
10. **Correct Answer: C**
**Maintaining cell shape** the cytoskeleton is a network of protein fibers (microtubules and filaments) that provides structural support and mechanical strength to the cell.

## COMPETETIVE CORNER

### SINGLE CORRECT OPTION TYPE

1. **Correct Answer: B**
Robert Hooke looked at thin slices of cork; since cork is dead plant tissue, the internal organelles had dried up, leaving only the empty "box-like" cell walls.
2. **Correct Answer: B**
Prokaryotic ribosomes are 70S (50S + 30S), while eukaryotic cytoplasmic ribosomes are 80S (60S + 40S).
3. **Correct Answer: C**
The nucleolus is the dense region within the nucleus where ribosomal RNA (rRNA) is synthesized and combined with proteins to form ribosome subunits.
4. **Correct Answer: C**
In an isotonic solution, the osmotic pressure is equal on both sides, so water enters and leaves the cell at the same rate.
5. **Correct Answer: B**
Hooke noted the cork looked like a honeycomb or the "cells" (small rooms) inhabited by monks.
6. **Correct Answer: B**
Unlike simple diffusion, facilitated diffusion requires specific transmembrane proteins (carriers or channels) to help polar molecules cross the bilayer without using energy.
7. **Correct Answer: C**
Proteins and lipids synthesized in the ER are packaged into transport vesicles that "bud off" and fuse with the *cis* face of the Golgi.
8. **Correct Answer: C**
Leucoplasts are non-pigmented plastids; examples include amyloplasts (starch), elaioplasts (oils), and proteinoplasts (proteins).
9. **Correct Answer: A**
Lignin is a complex polymer that provides extreme structural support to the secondary cell walls of woody plants and trees.
10. **Correct Answer: B**
The nuclear envelope consists of two concentric lipid bilayers (inner and outer membranes) perforated by nuclear pores.

### ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct Answer: A, C, D**
Cell Wall, Plastids, Large central vacuole
12. **Correct Answer: A, C, D**
Both provide mechanical support, transport materials, and increase surface area; only Smooth ER synthesizes lipids/glycogen.
13. **Correct Answer: B, D**
Mitochondria and Plastids/Chloroplasts are semi-autonomous and contain their own circular DNA

14. **Correct Answer: A, C, D**

    It covers the membrane, provides rigidity, and forms the middle lamella; protein synthesis happens in ribosomes

15. **Correct Answer: A, B, D**

    Prokaryotes have a nucleoid, 70S ribosomes, and microcompartments; they lack a complex cytoskeleton of microtubules/filaments

**ASSERTION AND REASON TYPE**

16. **Correct Answer: C**

    The membrane is fluid and dynamic, not rigid or static.

17. **Correct Answer: A**

    DNA contains the blueprints for proteins that run all cellular activities

18. **Correct Answer: A**

    The lack of a nucleus allows transcription and translation to happen simultaneously, speeding up growth and division.

19. **Correct Answer: A**

    Most microbodies (like peroxisomes) bud off from the specialized regions of the ER.

**MATRIX MATCHING TYPE**

20. **Correct Answer: A**

    a–S, b–R, c–Q, d–P

21. **Correct Answer: A**

    a–R, b–P, c–S, d–Q

**COMPREHENSION TYPE**

**Passage1**

22. **Correct Answer: C Cell**
23. **Correct Answer: C Virchow**
24. **Correct Answer: B Schleiden and Schwan**

**Passage2**

25. **Correct Answer: A Respiration**
26. **Correct Answer: B Powerhouse**
27. **Correct Answer: B Two**

## Previous Year Questions

1. **Correct Answer: B**

    All cells, regardless of type, must have a boundary (membrane), a metabolic floor (cytoplasm), and a way to make proteins (ribosomes).

2. **Correct Answer: A**

    Cytoplasm, Single, sap, Sugars, Salts

3. **Correct Answer: B**

    The cell wall is a non-living secretion of the protoplast.

4. **Correct Answer: D**

    Statement III is incorrect because microfilaments (actin) are essential for the membrane movement required to engulf particles.

5. **Correct Answer: B**

    The plasma membrane is **selectively permeable**, not impermeable.

6. **Correct Answer: D**

    Turgidity and rigidity are primarily functions of the **Vacuole** and **Cell Wall**, not the chloroplast.

7. **Correct Answer: D**

    This describes **plasmolysis**; water leaves the cell, causing the protoplast to shrink away from the cell wall.

8. **Correct Answer: D**

    Peroxisomes are **single-membrane** bound organelles.

# 2. Tissues

## Solutions

### Worksheet – 1

1. **Correct Answer: A**

   Histology is the branch of biology that focuses specifically on the microscopic structure of tissues.

2. **Correct Answer: C**

   Squamous epithelium It is called pavement epithelium because its flat, thin cells fit together like tiles on a floor.

3. **Correct Answer: B**

   The hair-like cilia move rhythmically to sweep dust and mucus out of the respiratory passages.

4. **Correct Answer: D**

   Sensory epithelium These cells are modified to receive external stimuli and transmit them to the nervous system.

5. **Correct Answer: B**

   Inner lining of the stomach and intestine These areas require glandular cells to secrete digestive juices, enzymes, and mucus.

6. **Correct Answer: C**

   Connective tissue It is the most widely distributed tissue, serving to bind, support, and protect various organs.

7. **Correct Answer: A**

   **Fat globules** Adipocytes store a large central droplet of fat, which physically displaces the nucleus to the edge of the cell.

8. **Correct Answer: D**

   **Tendon** Tendons are tough, inelastic cords of dense connective tissue that connect skeletal muscles to bones.

9. **Correct Answer: C**

   **To hold bones together at the joints** Ligaments are strong, elastic bands that connect one bone to another, ensuring joint stability.

10. **Correct Answer: B**

    **Elastic fibers** as the name suggests, these yellow fibers are composed of the protein elastin, which allows tissues to stretch and recoil.

### Worksheet – 2

1. **Correct Answer: B**

   Calcium phosphate is the primary inorganic mineral that provides hardness and rigidity to the bone matrix.

2. **Correct Answer: A**

   Hyaline cartilage has a smooth, glass-like bluish appearance and is typically covered by a perichondrium.

3. **Correct Answer: D**

   Erythrocytes (RBCs) circulate in the blood for about four months before being broken down.

4. **Correct Answer: C**

   Specifically, B-lymphocytes (a type of leucocyte) are responsible for producing antibodies to fight infections.

5. **Correct Answer: B**

   Platelets are not whole cells but small cytoplasmic fragments pinched off from large bone marrow cells called megakaryocytes.

6. **Correct Answer: C**

   Smooth muscle cells are fusiform (thick in the middle, tapered at the ends) and lack striations, making them involuntary.

7. **Correct Answer: D**

   These specialized junctions contain gap junctions that allow electrical impulses to spread rapidly for coordinated contraction.

8. **Correct Answer: B**

   The axon hillock is the cone-shaped area of the cell body (cyton) from which the axon extends.

9. **Correct Answer: A**

   The synapse is the functional junction where neurotransmitters are released to pass signals between neurons.

10. **Correct Answer: C**

    Spongy bone has a lattice-like structure where the "gaps" or pores are filled with red or yellow bone marrow.

## Worksheet-3

1. **Correct Answer: C**
   Meristematic cells are actively dividing and thus require a high nucleo-cytoplasmic ratio and dense cytoplasm to support rapid metabolic activity.
2. **Correct Answer: B**
   This meristem is located at the base of nodes or leaf blades in grasses, allowing the plant to regrow quickly after being grazed or mowed.
3. **Correct Answer: A**
   Differentiation is the developmental process where generalized meristematic cells undergo structural and functional changes to become specialized permanent cells.
4. **Correct Answer: C**

   Aerenchyma contains large air cavities that facilitate gas exchange and provide the buoyancy necessary for aquatic plants to float.

5. **Correct Answer: D**

   Collenchyma cells have localized wall thickenings made of pectin and cellulose, specifically at the corners where cells meet.

6. **Correct Answer: B**

   Unlike parenchyma and collenchyma which are living, sclerenchyma cells develop thick secondary walls impregnated with lignin and typically lose their protoplast at maturity.

7. **Correct Answer: C**

   Tracheids are the most primitive water-conducting elements, found in all vascular plants, whereas vessels are more specialized and primarily found in angiosperms.

8. **Correct Answer: A**

   Angiosperms are characterized by "porous wood" because they contain vessels; most gymnosperms lack vessels and are therefore termed "non-porous."

9. **Correct Answer: D**

   They are called sister cells because a single procambial cell divides asymmetrically to produce one sieve tube element and one companion cell.

10. **Correct Answer: B**

    While phloem parenchyma is common in dicots and gymnosperms, it is characteristically absent in the primary phloem of most monocotyledonous plants.

## Worksheet-4

1. **Correct Answer: D**

   Sachs classified tissues based on their location and function into the outermost covering, the main body fill, and the conducting strands.

2. **Correct Answer: C**

   While most dicots have kidney-shaped guard cells, grasses (monocots) possess specialized dumbbell-shaped cells to regulate stomatal opening.

3. **Correct Answer: B**

   Trichomes are epidermal hairs on the shoot that reflect light and create a boundary layer to reduce water loss through transpiration.

4. **Correct Answer: A**

   Dicot stems use flexible collenchyma for support in the hypodermis, while monocot stems utilize rigid, lignified sclerenchyma.

5. **Correct Answer: C**

   Casparian strips are waterproof bands found specifically in the endodermal cell walls to regulate the radial movement of water into the xylem.

6. **Correct Answer: B**

   Radial bundles are characteristic of roots, where xylem and phloem patches sit on different radii to facilitate efficient water absorption.

7. **Correct Answer: A**

   In dicot stems, vascular bundles are arranged in a precise ring (eustele), whereas in monocot stems, they are scattered throughout the ground tissue.

8. **Correct Answer: D**

   Passage cells are thin-walled endodermal cells located opposite the protoxylem that provide a low-resistance path for water to enter the vascular cylinder.

9. **Correct Answer: B**

   Secondary growth results from lateral meristems adding secondary vascular tissues and periderm, which increases the plant's diameter.

10. **Correct Answer: C**

    Monocot stem bundles have xylem and phloem on the same radius and lack cambium, meaning they cannot undergo secondary growth.

## Competitive Corner

**SINGLE CORRECT OPTION TYPE**

1. **Correct Answer: C**

   This tissue contains goblet cells that secrete mucus to trap particles and cilia that move the mucus out of the airways.

2. **Correct Answer: B**

   The extremely thin, single-layered squamous cells minimize the distance gases must travel between the air and blood.

3. **Correct Answer: D**

   Adipocytes shrink when they release their stored triglycerides (fat) into the bloodstream to be used as fuel.

4. **Correct Answer: C**

   The vertebral column serves as the central axis for the torso while housing and protecting the delicate spinal cord.

5. **Correct Answer: A**

   Dendrites are specialized branched projections that receive signals from other neurons and channel them toward the cyton.

6. **Correct Answer: B**

   Lateral meristems add layers of secondary xylem and phloem, causing the plant's diameter to expand over time.

7. **Correct Answer: C**

   Parenchyma is the most versatile and abundant ground tissue that fills the gaps between specialized tissue types.

8. **Correct Answer: D**

   Subsidiary cells support guard cells in the epidermis, while companion cells are specialized phloem cells that regulate sieve tube activity.

9. **Correct Answer: B**

   Also known as the rhizodermis, this layer produces root hairs to maximize the surface area for water absorption.

10. **Correct Answer: C**

    This lignified sheath provides essential mechanical support to the vascular bundles, which are scattered throughout the ground tissue.

## ONE OR MORE THAN ONE CORRECT OPTION TYPE

**11. Correct answer – C**

Sieve tubes, found in Angiosperms, form continuous longitudinal channels with sieve plates, making them more efficient at transporting food than the more primitive, individual sieve cells.

**12. Correct answer – A, B, D**

Parenchyma acts as filler, Collenchyma allows bending without breaking, and Chlorenchyma facilitates photosynthesis; however, C is incorrect because Sclerenchyma consists of dead cells with thick, lignified walls.

**13. Correct answer – A, B**

Apical meristems are specifically located at the tips of roots and shoots to facilitate primary growth and increase the length of the plant.

**14. Correct answer – A, B, C**

Meristematic tissues consist of undifferentiated, actively dividing cells that are responsible for the production of new cells throughout the plant's life.

**15. Correct answer – B, C, D**

Cardiac muscles are involuntary (unlike option A) and are uniquely characterized by intercalated discs and gap junctions that allow the myocardium to contract as a single, coordinated unit.

## ASSERTION AND REASON TYPE

**16. Correct Answer: A**

Because cells are so tightly packed together, there is no physical space for blood vessels to weave between them; they receive nutrients via diffusion from underlying tissues.

**17. Correct Answer: A**

Striations (stripes) are caused by the regular, repeating alignment of actin and myosin; without this alignment, the muscle looks "smooth."

**18. Correct Answer: C**

The assertion is correct, but the reason is entirely wrong: Sclerenchyma provides strength precisely because it is dead and has thick, lignified walls.

**19. Correct Answer: A**

In botany, "open" refers to the presence of a cambium (meristematic door) that allows the bundle to "open up" and produce new secondary tissue.

## MATRIX MATCHING TYPE

**20. Correct Answer: D**

a-R, b-S, c-Q, d-P, e-T

**21. Correct Answer: B**

a-T, b-R, c-P, d-Q, e-S

## COMPREHENSION TYPE

**Passage1**

**22. Correct Answer: A**

**23. Correct Answer: C**

**24. Correct Answer: D**

**Passage2**

**25. Correct Answer: B**

**26. Correct Answer: D**

**27. Correct Answer: A**

# 3. Diversity in Living Organisms

## Solutions

### Worksheet – 1

1. **Correct Answer: C**
   Aristotle was the first to classify living things into simple groups like "plants" and "animals" based on their physical appearance and habitat.
2. **Correct Answer: A**
   Linnaeus established the two-kingdom system, dividing all life into Kingdom *Plantae* (plants) and Kingdom *Animalia* (animals).
3. **Correct Answer: B**
   In 1866 (often cited in textbooks as the late 19th century/1866-1886 era), Haeckel introduced the third kingdom, *Protista*, to include microscopic organisms.
4. **Correct Answer: C**
   Whittaker expanded the system to five kingdoms (Monera, Protista, Fungi, Plantae, and Animalia) to better reflect cell structure and nutrition.
5. **Correct Answer: D**
   Classification is based on complex features like cell structure and how they get food, rather than just how big or small they are.
6. **Correct Answer: B**
   In binomial nomenclature, the first word (*Panthera*) represents the genus, and the second word (*leo*) represents the specific species.
7. **Correct Answer: D**
   In biological nomenclature, the term "Division" is used for plants, whereas "Phylum" is the equivalent term used for animals.
8. **Correct Answer: A**
   Phycology is the branch of botany dedicated to algae, while Mycology is the study of fungi.
9. **Correct Answer: C**
   Red algae store energy as Floridean starch, which is structurally very similar to glycogen and amylopectin.
10. **Correct Answer: B**
    Lichens are sensitive to sulfur dioxide and other toxins, acting as natural "bio-indicators" for air quality.

### Worksheet – 2

1. **Correct Answer: D**

   Double fertilization is a complex process unique to flowering plants where two sperm cells fertilize cells within the ovule.
2. **Correct Answer: B**

   The name "dicot" literally refers to the two cotyledons (seed leaves) present in the embryo of the seed.
3. **Correct Answer: C**

   Flowers are the reproductive organs of angiosperms where pollen (male) and ovules (female) are produced.
4. **Correct Answer: A**

   These features help gymnosperms, like pines, survive in dry or cold climates by preventing water from evaporating too quickly.
5. **Correct Answer: B**

   Cycas plants have a special partnership with blue-green algae (cyanobacteria) that live in their roots and provide nitrogen.
6. **Correct Answer: C**

   Polypodium is a common genus of ferns, which belong to the Pteridophyte group (unlike Pinus and Cycas which are gymnosperms).

7. **Correct Answer: B**
Unlike mosses, the visible, dominant stage of a fern (Pteridophyte) is the sporophyte, which has a double set of chromosomes.
8. **Correct Answer: C**
Pteridophytes were the first land plants to develop "plumbing" systems to transport water and food throughout the plant.
9. **Correct Answer: B**
The antheridium produces male gametes (sperm), while the archegonium is the female sex organ.
10. **Correct Answer: B**
In bryophytes (like mosses), the main green leafy part we see is the gametophyte stage, which produces gametes.

## Worksheet–3

1. **Correct Answer: B**
Invertebrates are defined by the absence of a notochord or vertebral column (backbone).
2. **Correct Answer: C**
The prefix "di-" means two, referring to the two germ layers (ectoderm and endoderm) present during development.
3. **Correct Answer: C**
Choanocytes use their flagella to create a water current that helps sponges filter food particles from the water.
4. **Correct Answer: B**
Physalia is a cnidarian because it possesses stinging cells called cnidoblasts used for defense and capturing prey.
5. **Correct Answer: B**
Ctenophores, commonly known as comb jellies, are famous for their ability to produce light (bioluminescence) in deep sea waters.
6. **Correct Answer: C**
Flame cells are primitive excretory organs that flicker like a flame as they filter waste and maintain water balance.
7. **Correct Answer: C**
Roundworms have a "false" body cavity because it is not fully lined by mesoderm tissue.
8. **Correct Answer: C**
Limulus is called a living fossil because it has remained virtually unchanged for millions of years.
9. **Correct Answer: B**
The radula acts like a tiny biological chainsaw or file that mollusks use to scrape food off surfaces
10. **Correct Answer: C**
The name Arthropoda literally means "jointed legs," which is their most recognizable physical feature.

## Worksheet –4

1. **Correct Answer: B**
**Echinodermata** – Adults have pentamerous radial symmetry, while larvae have bilateral symmetry.
2. **Correct Answer: C**
**Presence of an air bladder** – This organ provides buoyancy, allowing them to stay at depth without energy expenditure.
3. **Correct Answer: C**
**Proboscis gland** – This specialized gland is the primary excretory organ for hemichordates.
4. **Correct Answer: C**
**Only in the larval tail** – In Urochordata (Tunicates), the notochord is restricted to the tail and only during the larval stage.
5. **Correct Answer: A**

**Notochord is replaced by a vertebral column...** – All vertebrates start with a notochord, but it hardens into a backbone as they mature.

6. **Correct Answer: C**

**Cyclostomata** – These are primitive, jawless vertebrates like lampreys and hagfish.

7. **Correct Answer: B**

**Chondrichthyes** – This class includes sharks and rays, which lack an operculum (gill cover).

8. **Correct Answer: B**
**Reptilia** – While most reptiles have a 3-chambered heart, crocodiles are the first to show a 4-chambered heart.
9. **Correct Answer: D**
**Birds (Aves)** – Hollow bones reduce weight for flight, and the gizzard helps grind food.
10. **Correct Answer: C**
**Ornithorhynchus (Platypus)** – One of the few monotremes that lay eggs instead of giving birth to live young.

## Competitive Corner

### SINGLE CORRECT OPTION TYPE

1. **Correct Answer: C**
It failed to differentiate between prokaryotic and eukaryotic organisms.
2. **Correct Answer: B**
The science of classifying organisms.
3. **Correct Answer: C**
Kingdom (highest and broadest level).
4. **Correct Answer: C**
Primary endosperm nucleus (PEN).
5. **Correct Answer: B**
Enclosed in fruits.
6. **Correct Answer: B**
The spongocoel is the large central cavity of sponges where water enters through ostia and exits through the osculum.
7. **Correct Answer: B**
These organisms, commonly known as flatworms, lack a body cavity and are thin from the top (dorsal) to the bottom (ventral) surface.
8. **Correct Answer: B**
As the first vertebrates to move onto land, amphibians evolved these features to protect their eyes from drying out and to hear in an aerial environment.
9. **Correct Answer: C**
Struthio is the scientific genus for the ostrich, which belongs to the Ratite group of birds that have lost the ability to fly.
10. **Correct answer –B**
The Tornaria is a free-swimming, ciliated larva that is characteristic of the development of acorn worms (Hemichordates).

### ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Correct Answer: A, B, D**
Nematodes are usually dioecious, have internal fertilization, and vary in development.
12. **Correct Answer: A, C, D**
Reserve food is laminarin or mannitol, not glycogen
13. **Correct Answer: A, C**
Pinus has mycorrhiza; Cycas has coralloid roots
14. **Correct Answer: A, B, D**
Malpighian tubules are for excretion, not respiration
15. **Correct Answer: A, C, D**
The nerve cord in chordates is dorsal, not ventral

### ASSERTION AND REASON TYPE

17. **Correct Answer: B**
Both are true, but paired appendages (fins) do not explain why they aren't tetrapods (which requires four limbs)
18. **Correct Answer: A**
Both are true; the aquatic larval stage and external fertilization necessitate water.
19. **Correct Answer: A**
Pyrenoids are indeed proteinaceous storage bodies found in chloroplasts.
20. **Correct Answer: A**
The color variation is directly linked to the concentration of fucoxanthin.

### MATRIX MATCHING TYPE

21. **Correct Answer: C**
a-R, b-S, c-Q, d-P
22. **Correct Answer: A**
a-S, b-P, c-Q, d-R

## COMPREHENSION TYPE

**Passage1**

**23. Correct Answer: B**

**24. Correct Answer: C**

**25. Correct Answer: B**

**Passage2**

**26. Correct Answer: B**

**27. Correct Answer: B**

**28. Correct Answer: C**

## Previous Year Questions

**29. Correct Answer: D**

Bryophyta (2,3,4,5,6,7), Pteridophyta (4,5), Thallophyta (1,2,3,5,7,8).

**30. Correct Answer: D**

Only (iv) and (v) possess notochord.
(Note: (iii) Platyhelminthes do not have a notochord).

**31. Correct Answer: C**

Organism S is a "polyp" (Cnidaria), so its body is not differentiated into a visceral mass/foot (that's Mollusca).

**32. Correct Answer: D**

Y is the coelomic cavity lined by mesoderm (true coelomates).

# 4. Reproduction

## Solutions

### Worksheet – 1

1. **Correct Answer: (B) To produce new living organisms of the same kind**

   Explanation: Reproduction helps in producing new organisms of the same kind, ensuring continuity of species.

2. **Correct Answer: (A) Hypogynous**

   Explanation: In hypogynous flowers, the ovary is positioned above all other floral parts (superior ovary).

3. **Correct Answer: (C) Have genetic combinations from both parents**

   Explanation: Sexual reproduction combines genetic material from both parents, creating variation.

4. **Correct Answer: (B) Self-pollination**

   Explanation: Self-pollination occurs when pollen is transferred within the same flower.

5. **Correct Answer: (A) Dicots**

   Explanation: Dicots have two cotyledons (seed leaves) in their seeds.

6. **Correct Answer: (D) Pollen grains**

   Explanation: Pollen grains are involved in fertilization, not in germination.

7. **Correct Answer: (B) Large, bright-colored, and fragrant**

   Explanation: Bright colors and fragrance attract insects for pollination.

8. **Correct Answer: (C) Seed dispersal**

   Explanation: Seed dispersal is the process of seeds moving away from the parent plant.

9. **Correct Answer: (A) Transfer pollen grains to stigma**

   Explanation: Pollination involves transfer of pollen grains to the stigma.

10. **Correct Answer: (B) Male gamete and female gamete**
    Explanation: Fertilization is the fusion of male and female gametes to form a zygote.

### Worksheet-2

1. **Correct Answer: (B) Oviparous**
   Explanation: Oviparous animals lay eggs outside the mother's body, where development occurs.
2. **Correct Answer: (B) Two parents**
   Explanation: Sexual reproduction involves the fusion of male and female gametes from two parents.
3. **Correct Answer: (B) Testes**
   Explanation: Testes are the male reproductive organs that produce sperm.
4. **Correct Answer: (A) Mitochondria**
   Explanation: The middle piece of sperm contains mitochondria which provide energy for movement.
5. **Correct Answer: (C) Fallopian tube**
   Explanation: Fertilization in humans usually occurs in the fallopian tube.
6. **Correct Answer: (A) Ovum**
   Explanation: The ovum (egg) is the female gamete.
7. **Correct Answer: (C) 40 weeks**
   Explanation: Human pregnancy (gestation period) lasts about 40 weeks.
8. **Correct Answer: (C) 300 million**
   Explanation: Around 300 million sperms are released in a single ejaculation.
9. **Correct Answer: (A) Endometrium**
   Explanation: The endometrium is the inner lining of the uterus that thickens for implantation.
10. **Correct Answer: (B) Cervical canal**
    Explanation: The cavity present in the cervix is called the cervical canal.

## Worksheeet-3

1. **Correct Answer: (C) 28 days**

   Explanation: The average menstrual cycle in humans is about 28 days, though it may vary slightly.

2. **Correct Answer: (A) Days 1 – 5**

   Explanation: Menstruation (bleeding phase) usually occurs during the first 3 – 5 days of the cycle.

3. **Correct Answer: (B) Infections like UTIs**

   Explanation: Proper genital hygiene helps prevent infections such as urinary tract infections (UTIs).

4. **Correct Answer: (C) Sexual contact**

   Explanation: Sexually transmitted infections (STIs) mainly spread through sexual contact.

5. **Correct Answer: (A) Females**

   Explanation: Tubal ligation is a surgical method of contraception performed in females to block fallopian tubes.

6. **Correct Answer: (D) Couples facing infertility**

   Explanation: IVF is used to help couples who have difficulty conceiving naturally.

7. **Correct Answer: (C) Mature egg**

   Explanation: During ovulation, the ovary releases a mature egg (ovum).

8. **Correct Answer: (A) Frequent pregnancies**

   Explanation: Contraceptives help in spacing births and reducing health risks due to frequent pregnancies.

9. **Correct Answer: (B) In Vitro Fertilisation**

   Explanation: IVF means fertilisation of egg and sperm outside the body in a laboratory.

10. **Correct Answer: (C) STIs**

    Explanation: Barrier methods (like condoms) help prevent sexually transmitted infections.

## Competitive Corner

**SINGLE CORRECT OPTION TYPE**

1. **Correct Answer: (B) It ensures that the genetic lineage and species continue through new individuals.**
   Explanation: Reproduction does not make individuals immortal, but it allows the species to continue across generations, giving a sense of "group immortality."
2. **Correct Answer: (C) Condoms**
   Explanation: Condoms act as a physical barrier preventing sperm transfer and also protect against STIs.
3. **Correct Answer: (B) Menstruation will begin**
   Explanation: A sudden drop in estrogen and progesterone causes the uterine lining to shed, leading to menstruation.
4. **Correct Answer: (A) Foetus**
   Explanation: At the foetal stage, body parts are well developed and can be clearly identified.
5. **Correct Answer: (C) Uterus**
   Explanation: The zygote implants in the uterus, where it develops into an embryo and later a foetus.
6. **Correct Answer: (B) It reduces competition between the parent plant and the offspring for resources.**
   Explanation: Seed dispersal helps plants spread to new areas, avoiding overcrowding and competition.
7. **Correct Answer: (D) Semen**
   Explanation: Semen contains sperm along with nutrients like fructose and enzymes that help nourish and transport them.
8. **Correct Answer: (C) To allow them to float easily and be carried by water currents**
   Explanation: Light pollen helps aquatic plants in water pollination by enabling easy movement with currents.

9. **Correct Answer: (B) Professor Panchanan Maheshwari**
   Explanation: He is known as a pioneer of plant tissue culture research in India.
10. **Correct Answer: (C) Vasectomy**
    Explanation: Vasectomy is a surgical procedure in males where the vas deferens is cut to prevent sperm release

**ONE OR MORE THAN ONE CORRECT OPTION TYPE**

11. **Correct Answer: (A), (B)**
    Explanation: Testes produce sperms and secrete the hormone testosterone. They do not serve as a site for implantation, and sperm require a slightly lower temperature—not high temperature.
12. **Correct Answer: (A), (B), (D)**
    Explanation: Ovaries release ova, fertilisation occurs in the fallopian tube, and the uterus is where implantation and development occur. The vagina is not the site of foetal development.
13. **Correct Answer: (A), (B), (D)**
    Explanation: He contributed to plant tissue culture, helping produce disease-free plants and improving crop quality. He was not involved in vaccine discovery.
14. **Correct Answer: (A), (B), (C)**
    Explanation: Contraceptives help control population, prevent STIs (barrier methods), and reduce health risks by spacing pregnancies. They do not help in conceiving.
15. **Correct Answer: (A), (B), (C), (D)**
    Explanation: A complete flower contains all four whorls: sepals, petals, stamens, and pistil.

**ASSERTION AND REASON TYPE**

16. **Correct Answer: (A)**
    Explanation: Reproduction is essential for continuity of life because it replaces dead individuals with new ones, ensuring species survival.
17. **Correct Answer: (D)**
    Explanation: Fertilisation occurs in the fallopian tube, not the uterus. The reason is true (uterus is for implantation and development), but the assertion is false.
18. **Correct Answer: (A)**
    Explanation: His work in plant tissue culture helped agriculture by enabling production of disease-free and improved plants, correctly explaining the assertion.
19. **Correct Answer: (B)**
    Explanation: Both statements are true. The luteal phase involves the corpus luteum, and a drop in progesterone leads to menstruation. However, the reason explains menstruation, not specifically why the luteal phase is defined by corpus luteum.

**MATRIX MATCHING TYPE**

20. **Correct Answer: (A) a–Q, b–S, c–P, d–R**
21. **Correct Answer: (A) a–R, b–P, c–Q, d–S**

**COMPREHENSION TYPE**

**PASSAGE-1**

22. **Correct Answer: (C) Physical, mental, and social well-being**
23. **Correct Answer: (C) Toxic Shock Syndrome**
24. **Correct Answer: (B) Infections**

**PASSAGE-2**

25. **Correct Answer: (B) Prevent pregnancy**
26. **Correct Answer: (B) Barrier**
27. **Correct Answer: (C) Intrauterine device**

# 5. Sense Organs

## Solutions

### Worksheet – 1

1. **Correct Answer: D**
   Plato and Aristotle lived during the 4th century BC, which is roughly 2,300 years before our current time.
2. **Correct Answer: B**
   Albertus Magnus was a medieval scholar who, in the 13th century, identified that nerves are essential for carrying the sensation of touch.
3. **Correct Answer: C**
   Kepler was the astronomer who correctly identified that the eye functions like an optical instrument to focus light.
4. **Correct Answer: C**
   Modern science recognizes that our skin has different receptors specifically for feeling heat, cold, and physical force beyond simple "touch."
5. **Correct Answer: B**
   A stimulus is any change in the environment, such as light, sound, or heat, that triggers a reaction from our body.
6. **Correct Answer: C**
   The brain acts as the central processing unit that receives electrical signals from sense organs and interprets them as sights or sounds.
7. **Correct Answer: D**
   The iris is the muscular, pigmented ring that shrinks or expands to regulate how much light reaches the inner eye.
8. **Correct Answer: B**
   The sclera is the "white of the eye" and acts as a protective shield for the delicate internal structures.
9. **Correct Answer: C**
   The retina contains millions of photoreceptors that catch light to create the "picture" that is sent to the brain.
10. **Correct Answer: D**
    Rod cells are highly sensitive to low light levels, allowing us to see shapes and movement in the dark, unlike cone cells which need bright light for color.

### Worksheet – 2

1. **Correct Answer: C**
   These canals contain fluid that moves when you move your head, sending signals to the brain to help you stay upright.
2. **Correct Answer: D**
   The cochlea is the spiral-shaped part of the inner ear that turns mechanical sound vibrations into electrical impulses for the brain.
3. **Correct Answer: B**
   **Pinna -> Eardrum -> Ossicles -> Cochlea -> Auditory Nerve**
   Sound is collected by the outer ear (Pinna), vibrates the drum and bones (Ossicles), and is finally processed by the inner ear and nerve.
4. **Correct Answer: B**
   This type of permanent loss occurs when the sensitive hair cells or the auditory nerve itself are damaged and cannot send signals.
5. **Correct Answer: B**
   Earwax (cerumen) acts as a sticky trap to prevent dirt, debris, and small bugs from reaching and damaging the delicate eardrum.
6. **Correct Answer: B**
   The nose belongs to the sensory system for the sense of smell and the respiratory system for breathing air.
7. **Correct Answer: C**
   The nasal cavity acts like a natural air conditioner, preparing the air so it doesn't irritate the lungs.

8. **Correct Answer: B**
Sinuses are hollow spaces in the skull that can become swollen and painful when filled with fluid or infected.
9. **Correct Answer: D**
Inserting objects can cause bleeding, infections, or permanent damage to the sensitive nasal lining.
10. **Correct Answer: B**
These are non-cancerous teardrop-shaped growths that can sometimes make it difficult to breathe or smell.

## Worksheet 3

1. **Correct Answer: B**

Papillae are the small, visible bumps that house your taste buds and give the tongue its rough texture.

2. **Correct Answer: D**

Many poisonous plants and chemicals have a bitter taste, which serves as an evolutionary warning to stop us from eating them.

3. **Correct Answer: B**

Oral thrush is a fungal infection caused by Candida yeast, resulting in creamy white lesions on the tongue or inner cheeks.

4. **Correct Answer: B**

Extremely hot temperatures can burn the delicate papillae and temporarily damage your ability to taste.

5. **Correct Answer: C**

The hypodermis is made of fatty tissue that acts as an insulator to maintain body temperature and store fuel.

6. **Correct Answer: D**

Our skin synthesizes Vitamin D when it reacts to UV rays, which is essential for healthy bones and a strong immune system.

7. **Correct Answer: A**

Acne develops when sebum (oil) and dead skin cells clog the pores, often leading to inflammation or pimples.

8. **Correct Answer: C**
Vitiligo is a condition where the cells that produce melanin stop functioning, causing patches of skin to become much lighter.
9. **Correct Answer: B**
Proper wound care prevents bacteria from entering the body through breaks in the skin's protective barrier.
10. **Correct Answer: B**
The tongue constantly moves around the mouth during and after eating to sweep away food particles stuck between the teeth.

## Competitive Corner

**SINGLE OPTION CORRECT TYPE**

1. **Correct Answer: B**

This occurs when the eyeball is too long or the lens too curved, making distant light focus in front of the retina.

2. **Correct Answer: C**

Convex (converging) lenses help focus light onto the retina for people who otherwise see things focus behind it.

3. **Correct Answer: B**

These three tiny bones act as levers to increase the force of sound vibrations before they enter the fluid-filled inner ear.

4. **Correct Answer: A**

This tube connects the middle ear to the throat to balance internal pressure with the outside air.

5. **Correct Answer: B**

Epistaxis is the clinical term for bleeding from the blood vessels inside the nasal cavity.

6. **Correct Answer: B**

Sweat contains small amounts of metabolic wastes and excess salts, making skin a minor excretory organ.

7. **Correct Answer: B**
Melanoma is a dangerous form of cancer that begins in the melanocytes (pigment-producing cells) of the skin.
8. **Correct Answer: B**
Bitterness is an evolutionary safeguard that helps humans avoid consuming alkaloids and other poisons found in the wild.
9. **Correct Answer: C**
In myopia, the image of a distant object is formed in front of the retina, resulting in a blurry perception.
10. **Correct Answer: B**
Glossitis causes the papillae to disappear, leaving the tongue looking unusually red, smooth, and swollen.

**ONE OR MORE THAN ONE CORRECT OPTION TYPE**

11. **Correct Answer: A, B, D**
Senses protect us by identifying heat (acute responses), safe places (shelter), and hazards (environmental monitoring).
12. **Correct Answer: A, C, D**
Rods are for low-light vision and motion but do not provide the color or sharp detail that cone cells do.
13. **Correct Answer: A, C, D**
They work as a biological filter and humidifier to clean and prep air for the lungs.
14. **Correct Answer: A, B, C**
Glossitis, Thrush, and Geographic Tongue all change the visible surface of the tongue; Melanoma is a skin cancer.
15. **Correct Answer: A, C, D**
The skin manages heat via blood vessel changes and sweating, while sweat also removes salt and urea (excretion).

**ASSERTION AND REASON TYPE**

16. **Correct Answer: B**
Both statements are facts, but the correction for farsightedness does not explain why nearsightedness occurs.
17. **Correct Answer: D**
The retina, not the pupil, acts as the "cinema screen"; the pupil is just the hole that lets light in.
18. **Correct Answer: A**
The physical presence of hair and sticky mucus is exactly how the nose filters the air.
19. **Correct Answer: D**
The epidermis is actually the thinnest layer and contains no blood vessels; the dermis is the thick, vascular layer.

**MATRIX MATCHING TYPE**

20. **Correct Answer: A**
a-S, b-R, c-P, d-Q, e-T
21. **Correct Answer: B**
a-R, b-S, c-Q, d-T, e-P

**COMPREHENSION TYPE**

**Passage1**

22. **Correct Answer: B**
The nose detects chemicals (odors) floating in the air through olfactory receptors.
23. **Correct Answer: C**
The nose is the primary entry point for oxygen and exit for carbon dioxide in the breathing cycle.
24. **Correct Answer: B**
This wall of bone and cartilage keeps the two nostrils separate.

**Passage2**

25. **Correct Answer: B**
By surface area and weight, the skin is the largest organ of the human body.
26. **Correct Answer: B**
Its primary job is to act as a barrier against germs, injury, and dehydration.
27. **Correct Answer: B**
The skin is organized into the epidermis, dermis, and the deeper hypodermis.

# 6. Animal Behaviour

## Solutions

### Worksheet – 1

1. **Correct Answer: B**
   Ethology is the specific scientific study of animal behavior, typically focusing on behavior under natural conditions.
2. **Correct Answer: C**
   **A genetically programmed action present from birth** Innate behaviors are "hard-wired" into an organism's DNA and do not require prior experience or learning to be performed correctly.
3. **Correct Answer: D**

   **Instinct** A spider spinning a complex web perfectly the first time is a complex innate behavior known as an instinct.
4. **Correct Answer: B**

   The short stage in early life when imprinting occurs The sensitive (or critical) period is a specific developmental window where an animal is biologically primed to learn certain information, like identifying a parent.
5. **Correct Answer: C**

   Ivan Pavlov is famous for his experiments showing that dogs could be conditioned to salivate at the sound of a bell associated with food.
6. **Correct Answer: B**

   **Conditioning** This is a learned association where the animal connects a neutral stimulus (the cupboard opening) with a significant reward (food).
7. **Correct Answer: B**

   Imitation involves copying others; instinct is automatic and genetic Imitation is a learned behavior acquired by observing peers, whereas instinct is an inborn response that requires no observation.
8. **Correct Answer: A**
   How animals learn by watching and following adults Konrad Lorenz is best known for his work on **imprinting**, where young birds follow the first moving object they see during their sensitive period.
9. **Correct Answer: B**
   Albert Bandura Bandura's "Social Learning Theory" demonstrated that both humans and animals learn through observation, imitation, and modeling.
10. **Correct Answer: B**
    To interact with the environment to survive and continue the species Behavior is an evolutionary tool that allows animals to find food, avoid predators, and successfully mate to ensure the survival of their offspring.

### Worksheet – 2

1. **Correct Answer: B**
   A baby crying when hungry Crying is an inborn, unlearned response that ensures a human infant's needs are met by a caregiver.
2. **Correct Answer: C**
   The early emotional attachment between a child and caregivers In humans, imprinting is interpreted as the critical period for forming a deep psychological bond with parents or guardians.
3. **Correct Answer: B**
   **Conditioning** Feeling hungry at a specific time is a "conditioned response" where the body associates a neutral stimulus (the clock hitting 1:00 PM) with food.
4. **Correct Answer: C**

   To gather information about objects, food, or dangers Investigatory behavior is a survival mechanism used to assess whether a new element in the environment is a resource or a threat.

5. **Correct Answer: B**
To monitor migration routes and population size Scientists use physical or electronic tags to track animal movements across the globe and study their life cycles.
6. **Correct Answer: C**
It used boxes to climb or sticks to pull the banana closer This experiment demonstrated "insight learning," where an animal mentally rearranges objects to solve a physical problem.
7. **Correct Answer: B**
The direction and distance of a food source The waggle dance is a sophisticated form of symbolic communication used by bees to recruit others to a specific flower patch.
8. **Correct Answer: C**
Self-recognition and awareness Dolphins passing the mirror test suggests they possess "self-awareness," meaning they understand the reflection is themselves and not another animal.
9. **Correct Answer: C**
Through imitation by watching adults Social learning through imitation is the primary way human children acquire complex cultural traits like language and social etiquette.
10. **Correct Answer: B**
**Change their genetic code through curiosity** While intelligence allows animals to adapt their *behavior*, it cannot physically alter their underlying DNA or genetic code.

## COMPETITIVE CORNER

### SINGLE OPTION CORRECT TYPE

1. **Correct Answer: C**
Behavior is the total of all responses an organism makes to both internal cues (like hunger) and external stimuli (like a predator).
2. **Correct Answer: B**
While the other options involve learning or specific timing, the waggle dance is a genetically programmed, instinctual communication method.
3. **Correct Answer: C**
Unlike innate behavior, learned behavior is not fixed; it is modified based on experiences and environmental changes.
4. **Correct Answer: B**
Pavlov originally won the Nobel Prize in 1904 for his research on the digestive secretions of dogs, during which he discovered classical conditioning.
5. **Correct Answer: D**
Thorndike is famous for his "Puzzle Box" experiments with cats, which helped explain how animals learn through trial, error, and observation.
6. **Correct Answer: C**
The toddler is observing a specific motor skill from a peer/sibling and attempting to replicate it.
7. **Correct Answer: B**
The dance is the "language" used to translate spatial data; without it, the location of the food remains unknown to the hive.
8. **Correct Answer: C**
Imprinting must occur during a very specific "critical" or "sensitive" period early in life, or the bond will never form.
9. **Correct Answer: B**
Human "tagging" (IDs, digital tags) is social/administrative, whereas animal tagging is a tool for scientific data collection.
10. **Correct Answer: B**
These behaviors do not need to be taught; they are immediate biological reactions that prevent injury or ensure care.

### ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct Answer: B, D**
Genetically programmed and present from birth; flexibility and sensitive periods apply more to learned/imprinting behaviors
12. **Correct Answer: C, D**
Learning to avoid bad food and responding to a container are results of experience; the spider and duckling are driven by instinct/imprinting

13. **Correct Answer: A, C**
    Thorndike and Bandura focused on observational learning; Pavlov studied conditioning and Lorenz studied imprinting
14. **Correct Answer: A, B, D**
    Used for bird migration, digital social communication, and record-keeping
15. **Correct Answer: A, B, C, D**
    All are proven markers of high-level animal intelligence and cognitive ability

**ASSERTION AND REASNING TYPE**

16. **Correct Answer: A**

    Both A and R are true and R is correct explanation of A.
17. **Correct Answer: C**

    Reasoning (R) describes learned behavior, but the Assertion (A) is about innate behavior, which is rigid and fixed.
18. **Correct Answer: A**

    Both A and R are true and R is correct explanation of A.
19. **Correct Answer: D**

    The waggle dance is for communicating food location, not for mirror self-recognition.

**MATRIX MATCHING TYPE**

20. **Correct Answer: D**

    a-S, b-T, c-P, d-R, e-Q
21. **Correct Answer: C**

    a-R, b-P, c-S, d-T, e-Q

**COMPREHENSHION TYPE**

**Passage1**

22. **Correct Answer: B Present from birth**
23. **Correct Answer: C A spider spinning a web**
24. **Correct Answer: B DNA**

**Passage2**

25. **Correct Answer: C Biological and experiential factors**
26. **Correct Answer: B Instinct for survival**
27. **Correct Answer: A Reflex action (This is an automatic, involuntary survival instinct).**

# 7. NATURAL RESOURCES

## Solutions

### Worksheet – 1

1. **Correct Answer: (B) They are finite and may deplete**

   Explanation: Natural resources are limited, so using them wisely helps ensure availability for future generations.

2. **Correct Answer: (B) Non-renewable resources**

   Explanation: Fossil fuels take millions of years to form and cannot be replaced quickly once used.

3. **Correct Answer: (C) Pedology**

   Explanation: Pedology is the scientific study of soil formation, structure, and classification.

4. **Correct Answer: (A) Clay and loam**

   Explanation: Alluvial soil is rich in clay and loam, making it fertile and suitable for agriculture.

5. **Correct Answer: (B) Dissolving rock surface with acids**

   Explanation: Lichens secrete acids that break down rocks, helping in soil formation.

6. **Correct Answer: (C) Physical properties**

   Explanation: Texture, color, and porosity are observable physical characteristics of soil.

7. **Correct Answer: (B) Soil fertility**

   Explanation: Soil fertility refers to the soil's ability to provide nutrients for plant growth.

8. **Correct Answer: (D) Excess rocks**

   Explanation: Fertile soil should have nutrients, minerals, and organic matter—not excess rocks.

9. **Correct Answer: (B) Soil fertility maintenance**

   Explanation: Organic manure and crop rotation improve and maintain soil fertility.

10. **Correct Answer: (B) Chemical properties**

    Explanation: pH and salinity relate to the chemical nature of soil.

### Worksheet-2

1. **Correct Answer: (C) Vegetable peels**

   Explanation: Biodegradable wastes can be broken down by microorganisms; vegetable peels decompose naturally.

2. **Correct Answer: (B) Harmful substances accumulate and disturb soil composition**

   Explanation: Soil pollution occurs when toxic substances alter the natural balance of soil.

3. **Correct Answer: (D) Overgrazing and soil exposure**

   Explanation: Controlled grazing prevents removal of vegetation cover, protecting soil from erosion.

4. **Correct Answer: (B) Preventing movement of air and water**

   Explanation: Oil spills block soil pores, restricting air and water movement.

5. **Correct Answer: (A) Reduced fertility and productivity**

   Explanation: Pollution decreases soil quality, affecting plant growth.

6. **Correct Answer: (A) Roots hold soil particles and prevent erosion**

   Explanation: Tree roots bind soil, reducing erosion and improving soil stability.

7. **Correct Answer: (D) Collection, segregation, treatment, recycling, and safe disposal**

   Explanation: Proper waste management involves multiple steps to reduce environmental harm.

8. **Correct Answer: (B) Protecting soil and environment**

   Explanation: Recycling reduces waste accumulation and conserves natural resources.

9. **Correct Answer: (A) Tree roots hold soil particles together**

   Explanation: Afforestation prevents soil erosion by stabilizing soil with roots.

10. **Correct Answer: (B) Reduce soil pollution**

    Explanation: Reducing chemical use prevents harmful accumulation in soil.

## Worksheet-3

1. **Correct Answer: (B) Evaporation**

   Explanation: Evaporation is the process where liquid water changes into water vapour due to heat.

2. **Correct Answer: (C) Root nodules of leguminous plants**

   Explanation: Rhizobium bacteria fix nitrogen in the root nodules of leguminous plants.

3. **Correct Answer: (A) Carbon**

   Explanation: Carbon is the fundamental element forming the backbone of carbohydrates, proteins, and lipids.

4. **Correct Answer: (C) Respiration, burning, rusting, and decomposition**

   Explanation: Oxygen is involved in many processes like respiration, combustion, rusting, and decay.

5. **Correct Answer: (B) Absorb harmful UV radiation**

   Explanation: The ozone layer protects life on Earth by absorbing harmful ultraviolet radiation.

6. **Correct Answer: (D) 1987**
   Explanation: The Montreal Protocol was signed in 1987 to control substances that deplete the ozone layer.

7. **Correct Answer: (A) $CO_2$ utilization**
   Explanation: These organisms use carbon dioxide during photosynthesis.

8. **Correct Answer: (B) Release chlorine atoms that break down $O_3$ molecules**
   Explanation: CFCs release chlorine atoms which destroy ozone molecules.

9. **Correct Answer: (C) Oxygen utilization**
   Explanation: Rusting involves the reaction of iron with oxygen.

10. **Correct Answer: (C) Ammonification**
    Explanation: Ammonification converts organic nitrogen from dead organisms into ammonia.

## Competitive Corner

**SINGLE OPTION CORRECT TYPE**

1. **Correct Answer: (B) Because they can be depleted if the rate of consumption exceeds the natural rate of renewal.**

   Explanation: Forests and water are renewable, but overuse faster than their regeneration makes them exhaustible.

2. **Correct Answer: (C) By causing repeated expansion and contraction of rocks through temperature changes.**

   Explanation: Heating and cooling by the Sun cause rocks to expand and contract, leading to physical weathering.

3. **Correct Answer: (C) Chemical pesticides**

   Explanation: Non-biodegradable wastes like pesticides do not decompose easily and persist in the environment.

4. **Correct Answer: (A) Terrace farming**

   Explanation: Terrace farming reduces soil erosion by creating step-like structures on slopes.

5. **Correct Answer: (D) Rhizobium**

   Explanation: Rhizobium bacteria live in root nodules and fix atmospheric nitrogen.

6. **Correct Answer: (B) Because Earth is a closed system with a limited supply of these essential elements.**

   Explanation: Elements are reused through cycles because their supply on Earth is limited.

7. **Correct Answer: (C) Stratosphere**

   Explanation: The ozone layer is mainly present in the stratosphere.

8. **Correct Answer: (B) They grow in the fine particles created by lichens and add more organic matter when they die.**

   Explanation: Mosses help in soil formation by adding organic matter and improving soil quality.

9. **Correct Answer: (C) Oil spills**

   Explanation: Oil blocks soil pores, preventing air and water movement.

10. **Correct Answer: (C) Anabaena**

    Explanation: Anabaena (a blue-green alga) is commonly associated with paddy fields and fixes nitrogen.

**ONE OR MORE THAN ONE OPTION CORRECT TYPE**

11. **Correct Answer: (B), (C), (D)**

    Explanation: Biotic components include living organisms and organic matter like humus, bacteria, fungi, and nitrogen-fixing microbes. Mineral particles are abiotic.

12. **Correct Answer: (A), (B), (C)**

    Explanation: Lichens break rocks using acids, plant roots widen cracks, and microorganisms add humus. Animals do not prevent soil formation.

13. **Correct Answer: (B), (C), (D)**

    Explanation: Plastic, glass, and metals do not decompose naturally, unlike vegetable peels. Correct Answer: (A)

    Explanation: Afforestation helps prevent soil erosion because tree roots bind soil particles and stop them from being carried away by wind or water.

14. **Correct Answer: (A)**
    Explanation: Ozone layer depletion allows more harmful UV rays to reach Earth, increasing risks like skin cancer and cataracts. The reason correctly explains this.

15. **Correct Answer: (A)**
    Explanation: Dissolved oxygen is essential because aquatic animals depend on it for respiration.

16. **Correct Answer: (A)**
    Explanation: Plants cannot use atmospheric nitrogen directly; it must first be converted into usable forms like nitrates or ammonia by bacteria.

17. **Correct Answer: (A), (B), (C)**
    Explanation: Oxygen is used in respiration, combustion, and rusting, but released in photosynthesis.

18. **Correct Answer: (A), (C), (D)**
    Explanation: Ammonification, nitrification, and denitrification are key steps of the nitrogen cycle; photosynthesis is part of the carbon cycle.

**ASSERTION AND REASNING TYPE**

19. **Correct Answer: (A)**

    Explanation: Afforestation helps prevent soil erosion because tree roots bind soil particles and stop them from being carried away by wind or water.

20. **Correct Answer: (A)**

    Explanation: Ozone layer depletion allows more harmful UV rays to reach Earth, increasing risks like skin cancer and cataracts. The reason correctly explains this.

21. **Correct Answer: (A)**

    Explanation: Dissolved oxygen is essential because aquatic animals depend on it for respiration.

22. **Correct Answer: (A)**

    Explanation: Plants cannot use atmospheric nitrogen directly; it must first be converted into usable forms like nitrates or ammonia by bacteria.

## MATRIX MATCHING TYPE

23. **Correct Answer: (A) (a)–Q, (b)–R, (c)–S, (d)–P**
24. **Correct Answer: (A) (a)–R, (b)–P, (c)–S, (d)–Q**

## COMPREHENSION TYPE

**PASSAGE-1**

25. **Correct Answer: (A) Earthworms and microbes**
26. **Correct Answer: (B) Organic matter and microbial activity**
27. **Correct Answer: (C) Use of organic manure**

**PASSAGE-2**

28. **Correct Answer: (B) Conversion of nitrates into nitrogen gas**
29. **Correct Answer: (A) Ammonia to nitrites and nitrates**
30. **Correct Answer: (A) A nitrogen-fixing bacteria in soil**

## PREVIOUS YEAR QUESTIONS

1. **Correct Answer: (D) All of these**

   Explanation:

   Soil L (sandy) is not suitable for making pots (clayey soil is used).

   Soil M (loamy) is good for most plants, not xerophytes specifically.

   Soil N (clayey) is not best for garden plants due to poor aeration.

   Hence, all given statements are incorrect.

2. **Correct Answer: (D) Chlorofluorocarbons are the major ozone depleting substances which produce active chlorine in the presence of UV radiations.**

   Explanation: CFCs release chlorine under UV light, which destroys ozone. Other options are incorrect facts.

3. **Correct Answer: (A)**

   Explanation: DDT is non-biodegradable and accumulates in organisms, increasing concentration at higher trophic levels (biomagnification).

4. **Correct Answer: (B) Soil Y represents clayey soil and is suited for growth of crops like paddy.**

   Explanation:

   Soil X → sandy (low water holding)

   Soil Y → clayey (high water holding, poor aeration)

   Soil Z → loamy (balanced properties)

   Clayey soil is suitable for paddy.

5. **Correct Answer: (D) Soil L and N are not suitable for plant growth whereas soil M is best suited for plant growth.**

   Explanation:

   Soil L (sandy) → poor water retention

   Soil N (clayey) → poor aeration

   Soil M (loamy) → ideal balance for plant growth